

Incident Summary Report

Executive Summary

Analysis of the CIC-IDS2017 dataset revealed critical web attacks and infiltration events occurring on a Thursday. Traffic analysis exposed successful SQL injection data exfiltration alongside a reverse-shell infiltration that facilitated internal port scanning. The incident poses significant risks for live enterprise environments despite taking place within a synthetic testbed.

Environment Overview

The Canadian Institute for Cybersecurity provided the Intrusion Detection Evaluation Dataset containing over 7GB of PCAP files and labeled CSV flows. The infrastructure included:

- Victim Ubuntu web server (IP: 192.168.10.50)
- Windows Vista host (IP: 192.168.10.8)

The investigation utilized Wireshark for packet-level inspection and Splunk for indexing network telemetry logs.

Attack Timeline

Timeframe	Event Description
Thursday, 9:21 AM - 10:42 AM	Web attacks initiated through direct inbound connections from an external IP of 172.16.0.1.
Thursday, 2:28 PM - 3:46 PM	An internal Windows Vista host established a reverse shell connection to an external attacker IP of 205.174.165.73.

Indicators of Compromise (IOCs)

Telemetry flagged the following malicious activities:

- 1,507 brute-force attacks
- 652 cross-site scripting attempts
- 21 SQL injections
- 36 infiltration flows

Malicious activity targeted the `/dv/login.php` and `/dv/vulnerabilities/xss_r/` endpoints on the Ubuntu web server. Outbound connections over ports 444 and 4444 served as the external command channel for the reverse shell.

Impact Analysis

The operational impact remains contained since the event transpired within an isolated academic lab environment. Translating these tactics to a live enterprise network introduces critical severity due to the successful extraction of user database records. The compromised Windows Vista host acts as a pivot point for network-wide lateral movement.

Remediation Actions

Action Type	Recommendation
Web Applications	Deploy a web application firewall and enforce strict input sanitization to neutralize SQL injection and cross-site scripting vectors.
Network Defenses	Block unauthorized outbound connections on ports 444 and 4444 to sever reverse-shell communication channels. Segment the internal network.
Endpoint Protection	Deploy endpoint detection and response (EDR) solutions to contain compromised assets and prevent future lateral movement.